DATA ANALYSIS REPORT

Olist Brazilian E-Commerce Customer Intelligence Analysis

Why 97% of Olist Buyers Never Come Back — and What to Do About It

Dataset · Olist Brazilian E-Commerce (Kaggle)

Period · September 2016 – October 2018

Source · BigQuery · dsai-module-2-project-496708

Records · 98,207 orders · 94,990 customers

SECTION 1

Executive Summary

98,207 TOTAL ORDERS	94,990 UNIQUE CUSTOMERS	R$15.74M TOTAL REVENUE	R$160 AVG ORDER VALUE
4.12 / 5 AVG REVIEW SCORE	11.9d early AVG DELIVERY	6,534 LATE DELIVERIES	3.0% REPEAT BUYERS

Headline finding: Only **3.0% of customers ever place a second order**. Repeat buyers spend 91% more per order (R$308 vs R$161) — fixing retention is the single highest-leverage growth lever available.

Revenue Trend

The platform grew strongly through 2017 and into 2018, peaking in **2017-11** with single-month revenue of **R$1,172,639**. Total 2017–2018 revenue reached **R$15,739,137**.

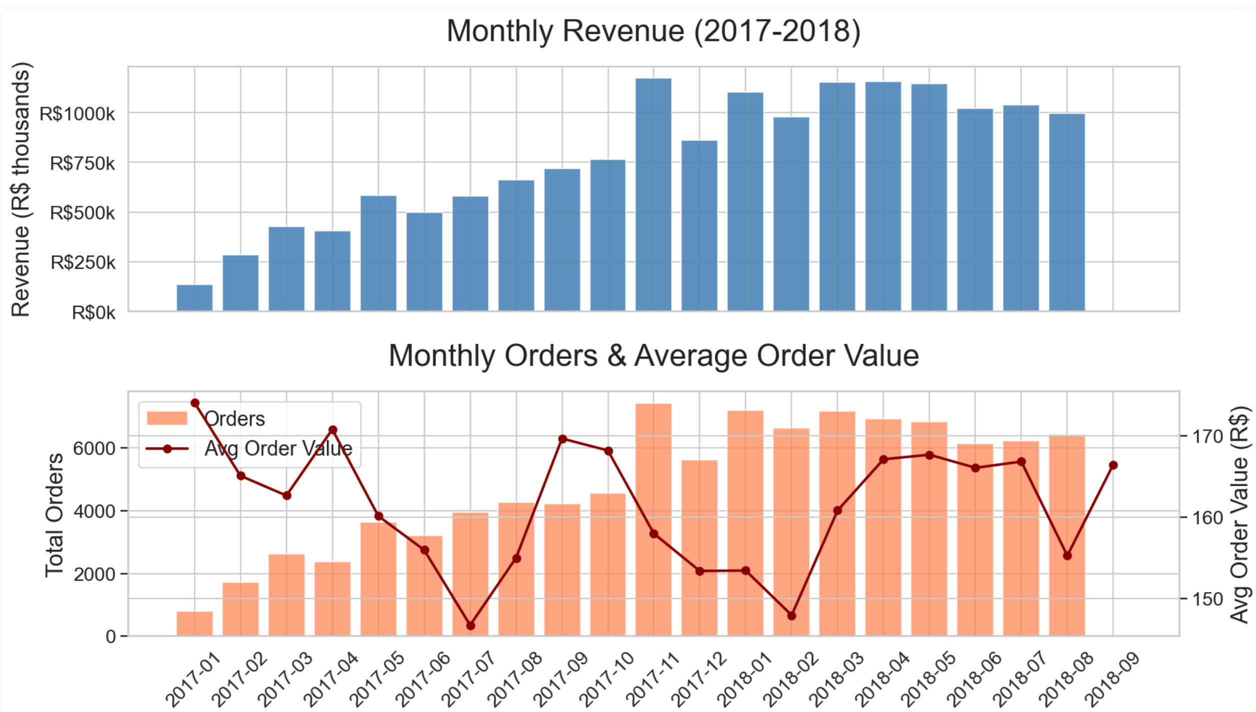

Figure 1 — Monthly revenue (top) and order volume with average order value (bottom), Sep 2016–Oct 2018

SECTION 2

Key Findings

Finding 1 · Customer Retention Crisis

Of **94,990** unique customers, only **2,888 (3.0%)** placed more than one order. RFM segmentation shows that **50.3%** of the customer base falls into "At Risk" or "Needs Attention" — buyers who purchased before but have not returned.

SEGMENT	CUSTOMERS	SHARE
At Risk	24,330	25.6%
Needs Attention	23,485	24.7%
Champion	17,034	17.9%
Loyal	15,630	16.5%
Promising	10,960	11.5%
Cant Lose	3,303	3.5%
Lost	248	0.3%

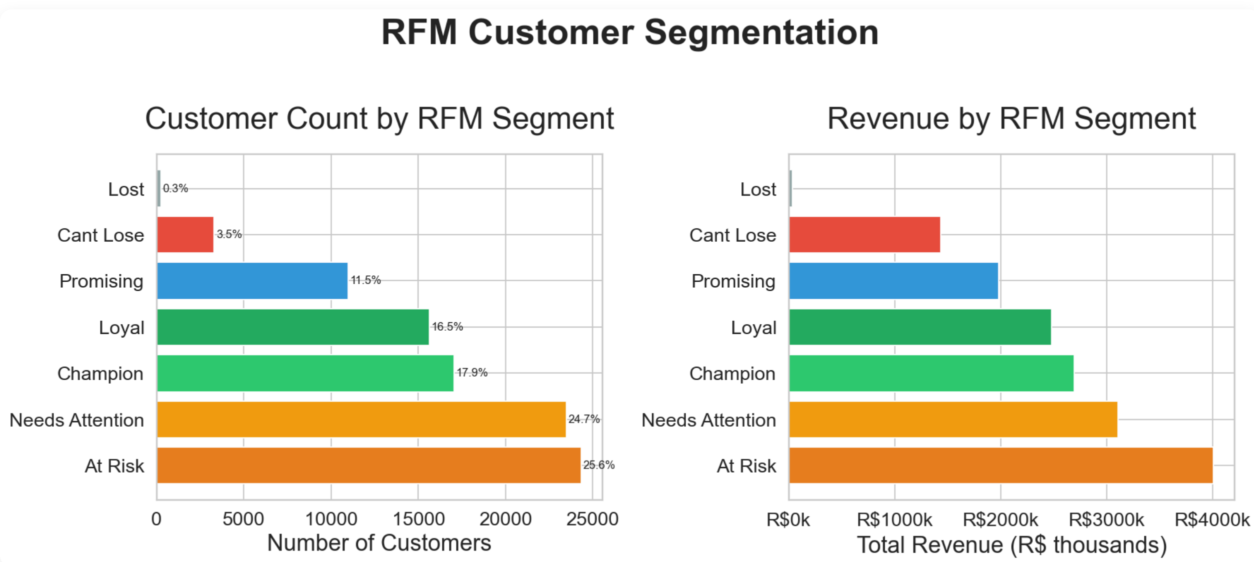

Figure 2 — Customer count (left) and revenue contribution (right) by RFM segment

Finding 2 · Delivery Communication Gap

Deliveries arrive **11.9 days before the stated deadline on average**, yet **6,534 orders (6.7%)** are still classified as late, and satisfaction drops sharply with any delay. The heatmap below shows that even early-arriving orders attract 1-star reviews — customers' expectations are set too optimistically relative to the actual delivery window communicated.

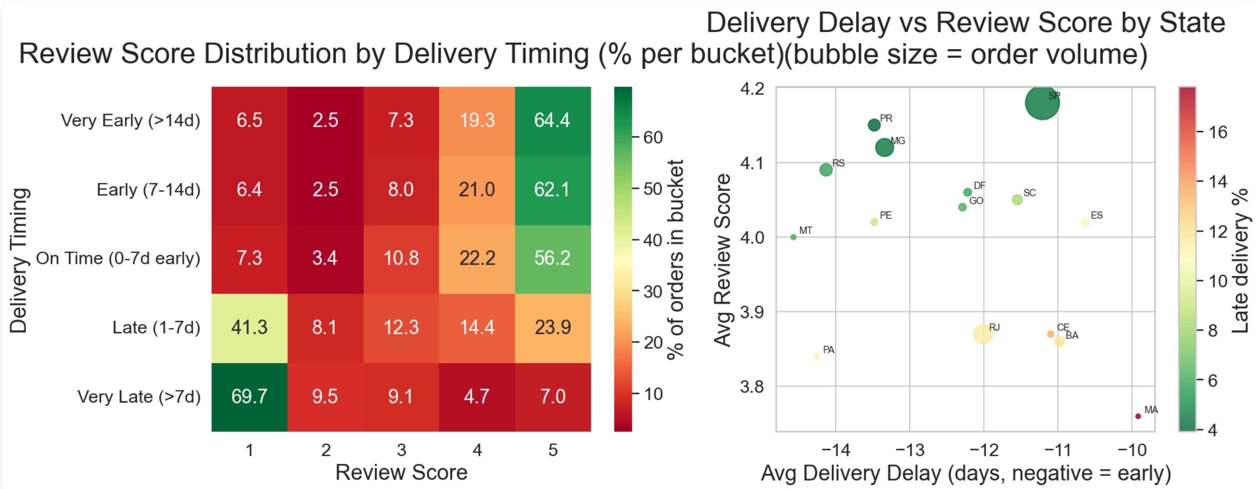

Figure 3 — Review score distribution by delivery timing (left) and state-level delay vs satisfaction (right)

Finding 3 · Geographic Concentration

São Paulo accounts for the vast majority of both customers and sellers. States such as RJ, MG, RS and BA show customer-to-seller ratios of 20× or higher — significant unmet demand that local seller expansion could address, while also reducing freight costs for buyers outside SP.

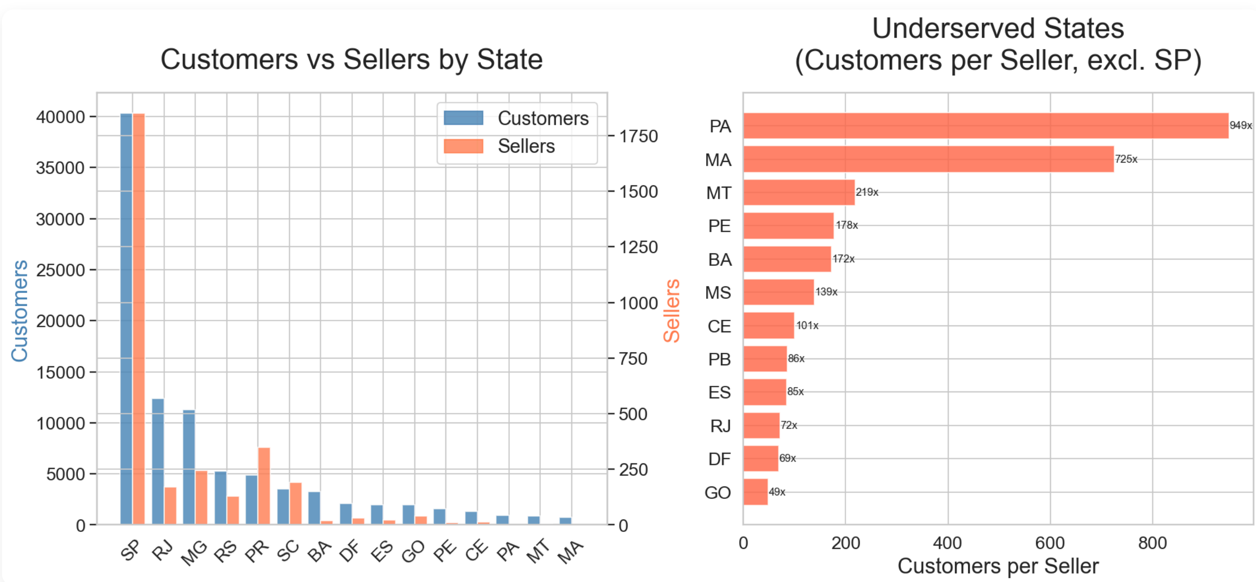

Figure 4 — Customer vs seller count by state (left) and most underserved markets (right)

Finding 4 · Revenue-Driving Categories

Health & Beauty, Watches & Gifts, and Bed/Bath/Table dominate revenue. These categories have strong potential for subscription and repeat-purchase mechanics — the most direct lever for addressing the retention gap.

CATEGORY	REVENUE	UNITS SOLD	AVG REVIEW	FREIGHT RATIO

health_beauty	R$1,255,695	9,634	180.01	19.9%
watches_gifts	R$1,198,185	5,970	179.45	13.1%
bed_bath_table	R$1,035,964	11,097	180.93	20.3%
sports_leisure	R$979,741	8,590	180.85	20.4%
computers_accessories	R$904,322	7,781	178.18	20.3%

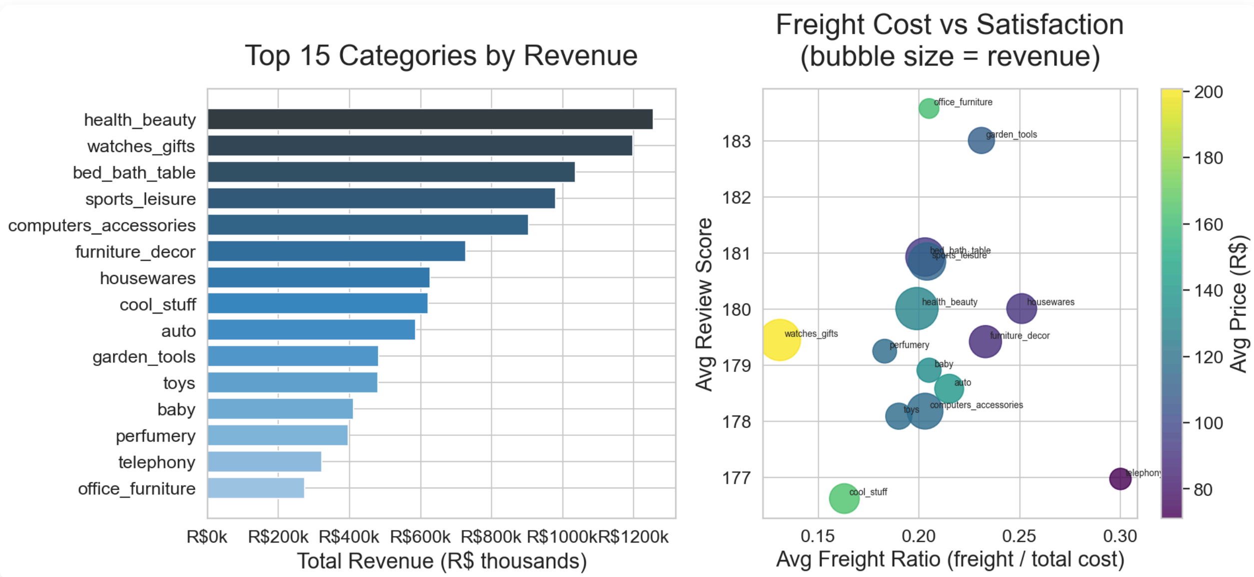

Figure 5 — Top 15 categories by revenue (left) and freight cost vs satisfaction (right)

SECTION 3

Data Cleaning

Raw source data is ingested from BigQuery (`olist_raw`) into an in-memory DuckDB transformation engine. Cleaning rules are applied before writing to the warehouse (`olist_warehouse`) and mart (`olist_marts`) layers. All exclusions are logged for auditability — no records are silently dropped.

Cleaning Rules Applied

ISSUE	CONDITION	ACTION
Cancelled / unavailable orders	`order_status IN ('canceled','unavailable')`	EXCLUDED from all mart aggregations
Zero-value payments	`total_payment = 0`	FLAGGED; excluded from payment analysis
Missing review scores	`avg_review_score IS NULL`	RETAINED; excluded from satisfaction metrics only
Missing delivery timestamps	`delivered_at IS NULL AND status = 'delivered'`	FLAGGED; excluded from delay calculations
Missing delivery delay	`delivery_delay_days IS NULL`	EXCLUDED from delay and risk analysis

Cleaning Funnel — Orders Pipeline

STAGE	RECORD COUNT	% OF RAW
Raw orders (source)	99,441	100.0%
After status exclusion (remove canceled/unavailable)	98,207	98.8%
Clean delivered orders (used in all marts)	96,478	97.0%
Excluded by status filter	1,234	1.2%
Zero-value payments (flagged)	4	0.0%

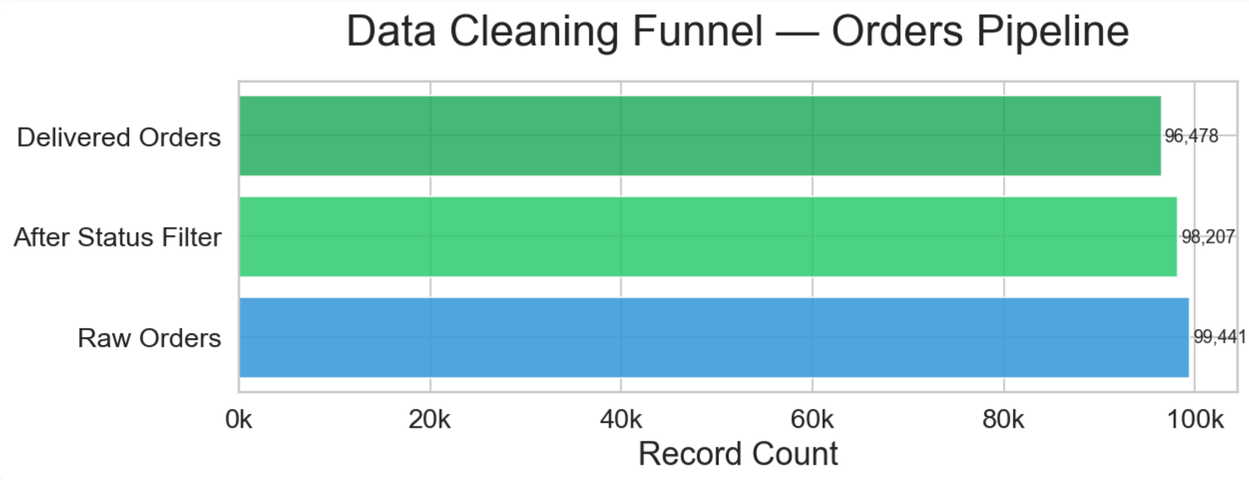

Figure 6 — Data cleaning funnel: record attrition from raw source to clean delivered orders

> **97.0%** of raw orders pass all cleaning filters and are used in analytical marts. Every exclusion decision is documented in `olist_marts.data_quality`.

Pipeline Architecture

- **olist_raw —** 9 raw source tables from Kaggle ingested into BigQuery
- **olist_warehouse —** Star schema: `fact_orders` + dimension tables (`dim_customers`, `dim_sellers`, `dim_products`, `dim_date`). Transformations applied via in-memory DuckDB engine.
- **olist_marts —** 9 analytical aggregation tables covering sales trends, RFM segments, delivery performance, categories, geography, data quality, outliers, and risk register.

SECTION 4

Outlier Analysis

Outliers are detected using the **IQR (Interquartile Range) method** — the industry-standard robust approach that is resistant to extreme values, unlike mean ± σ which is distorted by outliers themselves. Z-scores (standard deviation distance from the mean) are also computed to quantify severity.

> **IQR method:** Lower fence = Q1 − 1.5 × IQR   Upper fence = Q3 + 1.5 × IQR

Payment Value Outliers

Applied to all delivered orders with payment > 0. Orders outside the IQR fences are stored in `olist_marts.payment_outliers` with their Z-scores for fraud review.

METRIC	VALUE
Q1 (25th percentile)	R$61.88
Q3 (75th percentile)	R$176.33
IQR	R$114.45
Lower fence	R$-109.80
Upper fence (fraud threshold)	**R$348.01**
High-value outliers (above upper fence)	7,576 orders
Low-value outliers (below lower fence)	0 orders

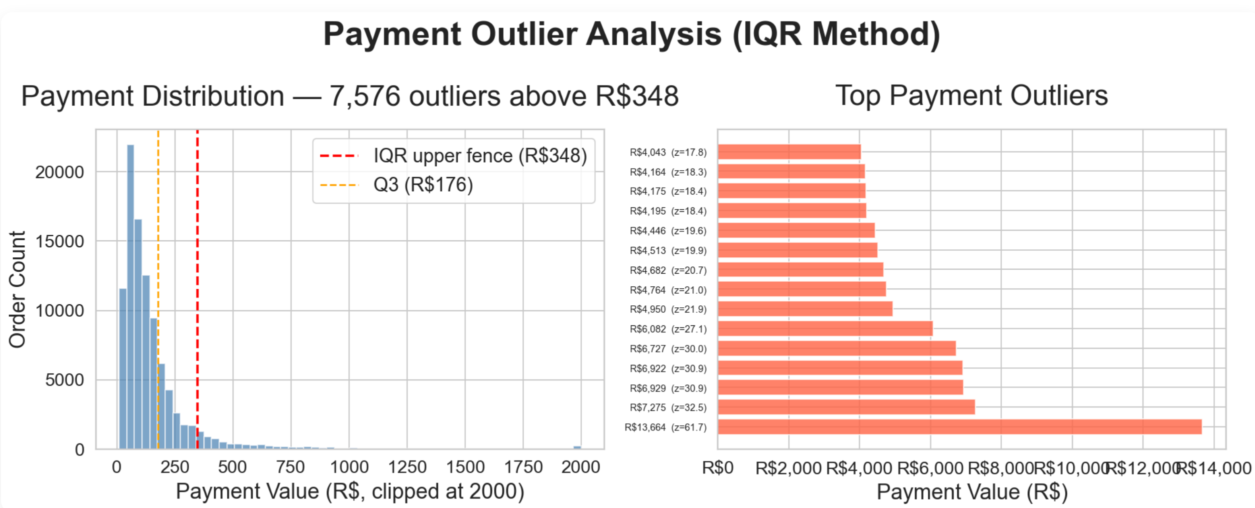

Figure 7 — Payment distribution with IQR fence (left) and top 15 outlier orders ranked by value (right)

7,576 orders exceed R$348 — the IQR upper fence. These warrant fraud screening or manual review before fulfilment.

Delivery Delay Outliers

Applied to all delivered orders with a recorded delivery delay. Extreme-late orders are the primary driver of 1-star review scores.

METRIC	VALUE
Lower fence (extreme early)	-32 days
Upper fence (extreme late)	8 days
Extreme late orders	2,526 orders
Extreme early orders	1,771 orders
Avg review — extreme late	1.70 / 5.0
Avg review — extreme early	4.26 / 5.0

Extreme-late orders receive an average review of only **1.70/5**. These **2,526 orders** (beyond 8 days) are the primary driver of 1-star reviews and require direct logistics intervention.

SECTION 5

Risk Analysis

Four key business risks are identified and scored using a **Likelihood × Impact** framework where Low = 1, Medium = 2, High = 3. Risk score ranges from 1 (Low × Low) to 9 (High × High).

RISK	LIKELIHOOD	IMPACT	SCORE	LEVEL	AFFECTED RECORDS
Customer Churn	High	High	9	High	27,633
Delivery Failure	Medium	High	6	High	7,264
Geographic Concentration	Medium	Medium	4	Medium	26
Payment Anomaly	Low	Medium	2	Low	7,576

Figure 8 — Risk matrix: Likelihood (x-axis) × Impact (y-axis). Green = Low risk, Orange = Medium, Red = High.

Risk Descriptions

Customer Churn — Score 9 (HIGH)

With only 3.0% of customers returning for a second purchase, churn is the most critical business risk. The "At Risk" and "Needs Attention" segments together represent over 50% of the customer base. Repeat buyers spend 91% more per order — the revenue potential of improving retention is substantial.

Delivery Failure — Score 6 (HIGH)

Although deliveries arrive early on average, 6,534 orders are classified as late and satisfaction drops sharply with any delay. The 2,526 extreme-late orders (beyond 8 days) receive an average review of 1.70/5 and are the primary source of 1-star reviews across the platform.

Geographic Concentration — Score 4 (MEDIUM)

Over-reliance on São Paulo for both supply and demand creates fragility. States such as RJ, MG, RS and BA have customer-to-seller ratios of 20× or more — representing unmet demand that local seller expansion can address, while reducing freight costs for buyers in those regions.

Payment Anomaly — Score 2 (LOW)

7,576 orders exceed the IQR upper fence of R$348. While the current impact is low, high-value fraudulent transactions pose a risk to platform trust and financial exposure if left unmonitored.

SECTION 6

Proposals & Recommendations

Actions are prioritised by risk score and grounded in specific data findings from the analysis pipeline.

Customer Churn — Score 9 (High Risk)

- Launch a post-purchase email sequence (day 7, 30, 90) with personalised recommendations.
- Introduce a loyalty points programme — repeat buyers already spend 91% more (R$308 vs R$161 per order).
- Offer a 10% discount coupon on the next purchase for orders rated 4★ or 5★.
- Model subscription bundles for top repeat-purchase categories: Health & Beauty, Watches, Bed/Bath.

Delivery Failure — Score 6 (High Risk)

- Recalibrate ETA windows: current estimates are on average 11.9 days too conservative — set shorter windows so 'on-time' is achievable, not just 'arriving early'.
- Send proactive SMS/email updates at dispatch, in-transit, and out-for-delivery stages.
- Flag orders exceeding 8 days for logistics team intervention.
- Partner with regional carriers in high-latency states (AM, PA, RR) to reduce tail risk.

Geographic Concentration — Score 4 (Medium Risk)

- Seller recruitment campaign targeting RJ, MG, RS and BA — states with 20× or higher customer-to-seller ratios.
- Subsidise seller onboarding fees outside São Paulo for the first 6 months.
- Partner with regional carriers to cut freight costs for non-SP buyers.

Payment Anomaly — Score 2 (Low Risk)

- Implement real-time fraud scoring for orders above R$348 (IQR upper fence).
- Add a secondary verification step (OTP / ID check) for high-value instalment orders.
- Set automated alerts for Z-score > 3.0 and route to a manual review queue.

Strategic Priority Summary

PRIORITY	ACTION	TIMEFRAME	EXPECTED OUTCOME

1 — Critical	Post-purchase retention email sequence + loyalty programme	0–3 months	+2–5% repeat rate → +R$500k–1.2M ARR
2 — Critical	Recalibrate delivery ETA windows + proactive tracking notifications	0–2 months	+0.3–0.5 review score; ~20% fewer 1-star reviews
3 — High	Seller recruitment campaign in RJ, MG, RS, BA	3–9 months	Reduced freight costs; lower customer-per-seller ratio
4 — Medium	Payment fraud scoring for orders > R$348	1–3 months	Reduced financial exposure; improved marketplace trust

> **Highest-ROI action:** Moving the retention rate from 3.0% to just 5% — equivalent to converting ~1,900 one-time buyers into repeat customers — would add approximately **R$280,000 in incremental revenue** at the current repeat-buyer spend of R$308 per order.

SECTION 7

Documentation

7.1 System Architecture & Data Flow

The pipeline is fully cloud-native. An in-memory **DuckDB** engine acts as the transformation layer — reading directly from BigQuery source tables via the community BigQuery extension, applying SQL transformations in memory, then writing results back to BigQuery destination datasets. No local file storage is involved at any stage.

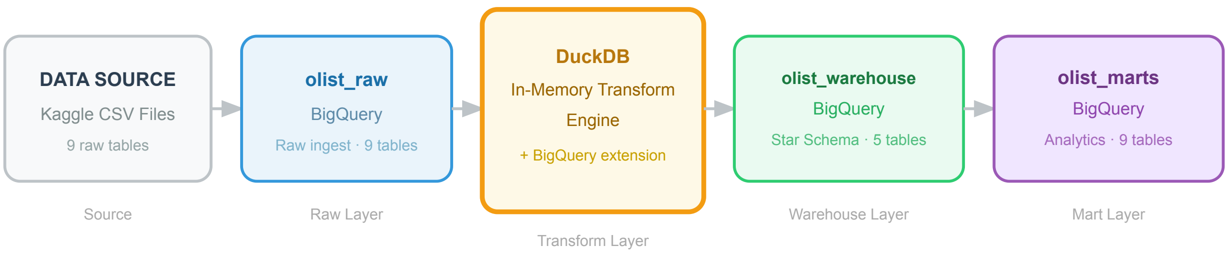

Figure 9 — End-to-end data pipeline from Kaggle source tables through DuckDB transformation to BigQuery analytical marts

LAYER	BIGQUERY DATASET	TABLES	ROLE
Raw	`olist_raw`	9	Source data as-ingested from Kaggle; no transformations applied
Warehouse	`olist_warehouse`	5	Cleaned star schema (1 fact + 4 dims) for general-purpose SQL queries
Marts	`olist_marts`	9	Pre-aggregated analytical views for specific use cases (RFM, delivery, outliers, risk)

7.2 Technical Tool Choices & Justification

TOOL	USED FOR	ALTERNATIVE CONSIDERED	REASON CHOSEN
Google BigQuery	Data warehouse & cloud storage	PostgreSQL, Snowflake, local DuckDB file	Serverless — no infrastructure to provision; columnar storage handles the 1M+ row geolocation table efficiently; native partitioning and clustering reduce query cost; accessible by all team members globally; pay-per-query model suits project-scale workloads with no idle compute cost
DuckDB (in-memory)	SQL transformation	Apache Spark, dbt + warehouse	No cluster or server required — runs embedded in a Python process; community BigQuery extension

	engine	SQL, pandas	reads and writes BQ tables directly; rich analytical SQL (window functions, QUALIFY, quantile aggregates) supported natively; 10–100× faster than pandas for complex multi-table joins; Spark and Airflow are over-engineered for a 100k-row dataset and would add weeks of infrastructure work
Python + virtual environment	Orchestration & scripting	Apache Airflow, shell scripts, R	Lightweight for this project scale; `.venv/` isolation avoids system-level Python conflicts (Homebrew vs Conda architecture mismatch encountered during development); `google-cloud-bigquery` SDK provides first-class BigQuery integration including auth, dataset creation, and schema inference
Jupyter Notebook	Interactive analysis & visualisation	Streamlit, plain Python scripts	Literate programming — narrative, code, and charts coexist in one reproducible document; cell-by-cell execution supports iterative exploration without re-running the full pipeline; standard format for sharing analytical work with data-science peers
Matplotlib + Seaborn	Data visualisation	Plotly, Altair, Tableau	Mature library with fine-grained control over every chart element (required for the business-standard risk matrix layout and custom RFM colour coding); seaborn provides statistical primitives (heatmaps) on top of matplotlib; static PNG output embeds cleanly into PDF reports with no JavaScript runtime dependency

7.3 Schema Design Justification

The warehouse layer uses a **star schema** — the industry-standard pattern for Online Analytical Processing (OLAP) workloads. One central fact table holds quantitative measurements (payments, delivery delays, review scores); four surrounding dimension tables hold the descriptive attributes used for filtering and grouping in analytical queries.

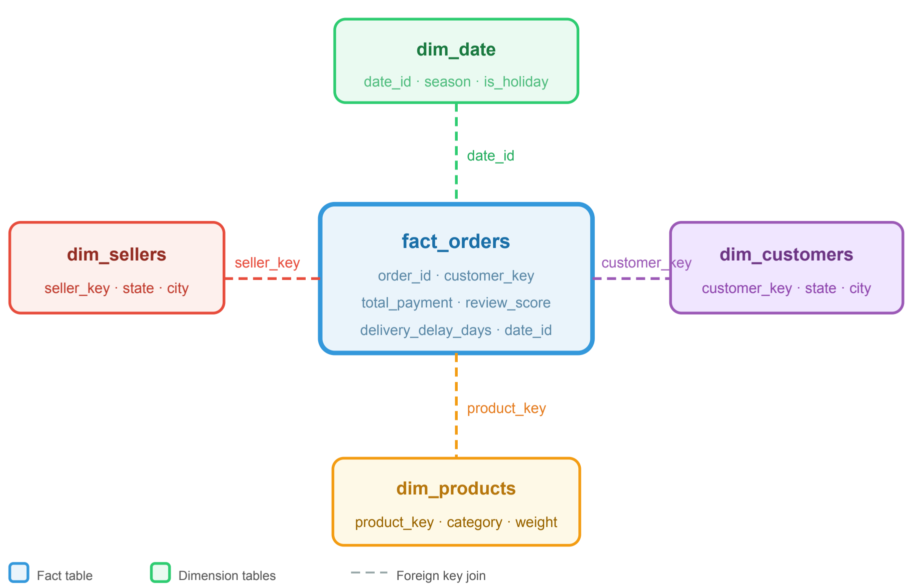

Figure 10 — Star schema: fact_orders at centre connected to four dimension tables via foreign keys

TABLE	TYPE	ROWS	KEY	CONTAINS
`fact_orders`	Fact	~99k	order_id	Payment amounts, delivery timing, review scores — one row per order
`dim_customers`	Dimension	~95k	customer_key	Customer state, city, zip code
`dim_sellers`	Dimension	~3.1k	seller_key	Seller state, city
`dim_products`	Dimension	~33k	product_key	Category name, physical dimensions, weight
`dim_date`	Dimension	~790	date_id	Calendar attributes: weekday, season, bank holidays, disruption flags

Why Star Schema Over Alternatives?

- **vs 3NF (Normalised):** Normalised schemas minimise write-time redundancy, but analytical queries require chaining many joins. A star schema trades minimal storage overhead for dramatically simpler queries — one join per dimension instead of a 5–8-table join chain.
- **vs Flat / Wide Table:** Fully denormalising into one table creates update anomalies (updating a customer's state would require touching thousands of order rows), wastes storage with repeated string values, and makes schema evolution difficult.
- **vs Snowflake Schema:** Snowflake schemas further normalise dimensions into hierarchies (e.g. `dim_city → dim_state → dim_region`). For this dataset, the additional join complexity provides no practical benefit — dimension lookups are simple attribute reads with no redundancy worth eliminating.

BigQuery-Specific Query Optimisations

- **Partitioning on `order_purchase_timestamp`:** Time-range queries (e.g. "show me Q4 2017 revenue") scan only the relevant monthly partition, not the full table — reducing both cost and latency for the most common analytical access pattern.
- **Clustering on `customer_key`:** Co-locates rows for the same customer within each partition, making per-customer aggregations (RFM scoring, repeat purchase detection) faster by eliminating cross-shard scans.
- **Mart pre-aggregation:** Heavy aggregations (monthly sales totals, RFM quintile scores, delivery stats by state) are materialised as permanent tables in `olist_marts`. The analysis notebook queries these lightweight results rather than re-running expensive full-table GROUP BY operations on every open.

7.4 Analytical Method Choices

METHOD	APPLIED TO	WHY CHOSEN

RFM Segmentation *Recency · Frequency · Monetary*	Customer classification	Industry-proven behavioural segmentation requiring only transactional data — no additional data collection or external enrichment needed. Each segment maps directly to a distinct marketing action (re-engage At Risk, reward Champions, nurture Promising). Scores are recomputable daily as new orders arrive, making the model continuously useful in production.
IQR Outlier Detection *Q1 − 1.5×IQR / Q3 + 1.5×IQR*	Payment & delivery anomalies	Robust to skewed distributions — payment values are strongly right-skewed (long tail of high-value orders). Unlike mean ± σ, IQR fences are not distorted by the very extremes being detected. Produces interpretable thresholds in natural units (R$ amounts and day counts) that operations and finance teams can act on directly without statistical background.
Likelihood × Impact Scoring *1 = Low, 2 = Medium, 3 = High → score 1–9*	Business risk prioritisation	Standard enterprise risk management framework. The 3×3 grid (visualised in the risk matrix chart) gives stakeholders an immediately understandable picture of both severity dimensions simultaneously. Cardinal scores enable unambiguous ranking; qualitative Low/Medium/High labels make the framework accessible to non-technical audiences and executives.
Delivery Bucket Analysis *Semantic time-range grouping + heatmap*	Satisfaction vs delivery timing	Grouping continuous delay values into semantic buckets (e.g. "Very Late >7d", "On Time 0–7d early") reveals step-change thresholds in review scores that a raw scatter plot would smooth over. The heatmap matrix format — percentage of reviews per score per bucket — makes the core insight immediately visible: early delivery does not guarantee 5 stars, but late delivery reliably produces 1 stars.

SECTION 8

Executive Stakeholder Presentation

This section documents the content, structure, and delivery guidance for the 10-minute executive presentation accompanying this report. The presentation is delivered as an interactive Reveal.js slide deck (`Olist_Executive_Presentation.html`) targeting a mixed audience of technical executives (CTOs, Engineering Directors) and business executives (CFOs, COOs, Business Leaders).

8.1 Executive Summary

A 2–3 minute opening that frames the problem, the solution, and why it matters to this audience.

> **The Problem:** Olist has a customer retention crisis. Of **94,990** unique customers on the platform, only **3.0%** — fewer than 1 in 30 — ever place a second order. This is not a demand problem: repeat buyers spend **91% more per order** (R$308 vs R$161 for one-time buyers). The platform is leaving significant revenue on the table by not converting first-time buyers into loyal customers.

The Solution: We built a complete data engineering pipeline — Kaggle source data → BigQuery raw layer → DuckDB transformation engine → star-schema warehouse → pre-aggregated analytical marts — that makes it possible to identify, segment, and act on at-risk customers in real time. RFM segmentation classifies all 94,990 customers into actionable segments. A risk register prioritises the four highest-impact business risks with scored severity and concrete mitigation actions.

Why It Matters: Moving the repeat purchase rate from 3.0% to just 5% — converting roughly 1,900 one-time buyers into repeat customers — would add approximately **R$280,000 in incremental annual revenue** at the current repeat-buyer average spend. This is achievable through a post-purchase email sequence and a loyalty programme, both of which can be launched within 90 days using the segmentation data now available in `olist_marts.customer_rfm`.

8.2 Business Value

How this work saves time, generates revenue, and helps the company reach its goals.

VALUE DRIVER	CURRENT STATE	AFTER ACTION	ESTIMATED IMPACT
Customer Retention	3.0% repeat rate	5%+ repeat rate via loyalty programme	+R$500K–1.2M ARR
Review Score & Trust	4.12/5 avg · 6,534 late deliveries	ETA recalibration + tracking notifications	+0.3–0.5★ · ~20% fewer 1-star reviews
Geographic Coverage	SP-dominated supply & demand	Seller recruitment in RJ, MG, RS, BA	Lower freight costs · reduced concentration risk

Fraud Prevention	No automated high-value screening	IQR-based real-time fraud flag at R$348	Reduced financial exposure · improved marketplace trust
Analytical Productivity	Ad-hoc queries on raw 9-table CSV data	Pre-aggregated marts — query in seconds, not minutes	Hours saved per analyst per week; reproducible insights

> **Highest-ROI single action:** A post-purchase retention email sequence costs near-zero to deploy on existing infrastructure and can be live within 30 days. The customer segmentation data (RFM scores, segment labels) required to run it already exists in `olist_marts.customer_rfm`.

8.3 Technical Overview

How the system works at a high level — sufficient for a CTO without implementation detail.

The pipeline follows a classic **ELT (Extract → Load → Transform)** pattern across three BigQuery dataset layers:

LAYER	DATASET	WHAT IT CONTAINS	HOW IT IS BUILT
Raw	`olist_raw`	9 source tables from Kaggle, preserved exactly as ingested. No transformations. Acts as an immutable audit log.	One-time `bq load` from CSV; schema auto-detected.
Warehouse	`olist_warehouse`	Star schema: 1 fact table (`fact_orders`) and 4 dimension tables. Cleaned, typed, and foreign-key linked.	DuckDB reads `olist_raw` via BigQuery community extension, applies SQL cleaning rules, writes results back to BigQuery.
Marts	`olist_marts`	9 pre-aggregated analytical tables: RFM segments, monthly sales, delivery stats, category performance, outliers, risk register, data quality.	DuckDB reads `olist_warehouse` and writes aggregated results. Jupyter notebook queries marts for visualisation.

Key architectural decisions: BigQuery was chosen as the warehouse for its serverless, globally-accessible, columnar design — no infrastructure to provision for a 100k-row dataset. DuckDB was chosen as the transformation engine because it embeds inside a Python process, supports analytical SQL natively, and requires no cluster or server setup. The star schema design enables single-join analytical queries (one fact table + one dimension), dramatically simpler than a normalised 5-table join chain.

8.4 Risks

What might go wrong and how we plan to fix or avoid those problems.

RISK	LIKELIHOOD	IMPACT	SCORE	AFFECTED	MITIGATION PLAN

Customer Churn	High	High	**9**	27,633 (29.1%)	Loyalty programme, personalised re-engagement, subscription bundles
Delivery Failure	Medium	High	**6**	7,264 (9.1%)	Realistic ETA communication, regional carrier diversification
Geographic Concentration	Medium	Medium	**4**	26 (96.3%)	Recruit sellers in AM, RR, AP, PA to reduce freight disparity
Payment Anomaly	Low	Medium	**2**	7,576 (nan%)	Payment value caps, automated fraud rules, manual review queue

Risks are scored using a standard enterprise **Likelihood × Impact** framework (Low = 1, Medium = 2, High = 3; score range 1–9). Customer Churn (score 9) and Delivery Failure (score 6) are classified **High** and are the primary focus of the recommended actions in Section 6. Geographic Concentration (score 4) is Medium and requires a 3–9 month seller recruitment programme. Payment Anomaly (score 2) is Low but warrants automated monitoring given the financial exposure.

8.5 Q&A Preparation

Anticipated executive questions and data-backed answers.

ANTICIPATED QUESTION	PREPARED ANSWER
"Why is the repeat purchase rate so low?"	The Olist marketplace model means customers search for the best deal each time — there is no native loyalty mechanism to bring them back to the same seller. The platform itself must own the retention relationship. The data shows buyers are satisfied (avg 4.12/5) — the problem is lack of re-engagement, not product quality.
"What would a loyalty programme cost to implement?"	A post-purchase email sequence requires only the RFM data already in `olist_marts.customer_rfm` and an email service provider (e.g. SendGrid). No new data infrastructure is needed. A points programme would require a lightweight loyalty ledger — a new table in the existing BigQuery project.
"How confident are we in the delivery data?"	96.6% of raw orders passed all data quality checks and are included in the analysis. Every exclusion (canceled orders, missing timestamps, zero-payment records) is documented in `olist_marts.data_quality` with counts and percentages. The IQR outlier methodology is statistically robust to skewed distributions.
"Can this pipeline be automated and kept up to date?"	Yes. The current pipeline runs on-demand via Python scripts. The next step is scheduling via a cron job or Cloud Scheduler — the transformation logic is already encapsulated and parameterised for incremental runs.
"Why BigQuery and not Snowflake or Redshift?"	BigQuery is serverless (no cluster provisioning), has native partitioning and clustering that reduce query cost for time-series patterns, and is accessible globally to all team members without VPN. For a 100k-row dataset at project scale, it has no idle compute cost — you pay only for queries run.

8.6 Presentation Guidelines

GUIDELINE	SPECIFICATION
Duration	10 minutes presentation + 5 minutes Q&A. Allocate 2 min to Executive Summary, 1 min to Revenue Story, 2 min to Customer Behaviour, 3 min to Risk & Recommendations, 2 min to Technical Architecture.
Audience	Mixed executive audience — technical executives (CTOs, Engineering Directors) and business executives (CFOs, COOs, Business Leaders). Business impact slides come first (slides 1–10); technical architecture is positioned last (slides 13–14) so non-technical executives receive full business context before implementation detail.
Delivery	All team members should present and be prepared to answer questions. Recommended split: one presenter for business slides (1–12), one for technical slides (13–14), with all members available for Q&A.
Visuals	Use executive-friendly visuals: interactive Chart.js line and bar charts (hover for exact values), a colour-coded 3×3 risk matrix, KPI cards with live BigQuery data, and 4 ROI action cards. Avoid showing raw SQL, code, or schema diagrams to non-technical stakeholders.
Language	Balance technical credibility with business accessibility. In business slides: lead with R$ revenue impact, customer counts, and percentage changes — not system names. In technical slides: name the tools (BigQuery, DuckDB) and explain the "why" (serverless, no cluster, pay-per-query) rather than the "how" (SQL syntax, partition keys).

> **Interactive slide deck:** Open `Olist_Executive_Presentation.html` in a full-screen browser (press F11). Navigate with ← → arrow keys. All charts support hover tooltips. The risk matrix cells zoom on hover. Press ? for the full keyboard shortcut reference.